

Tab 1



MERN Stack Development Notes

This document provides a structured set of notes for the MERN (MongoDB, Express.js, React, Node.js) stack, outlining key concepts and commands for full-stack web development.

1. MongoDB (Database)

MongoDB is a document database designed for ease of application development and scaling.

Key Concepts

- **Document:** A record in MongoDB, analogous to a row in a relational database, but structured as a JSON-like BSON object.
- **Collection:** A group of MongoDB documents, analogous to a table.
- **Schema:** Defined using Mongoose, it structures the data within a collection.
- **Mongoose:** An Object Data Modeling (ODM) library for MongoDB and Node.js.

Basic Commands

Action	MongoDB Shell Command	Description
Connect to DB	<code>mongo</code>	Starts the MongoDB shell.
Use/Create DB	<code>use <database_name></code>	Switches to a database or creates it if it doesn't exist.

Action	MongoDB Shell Command	Description
Show DBs	<code>show dbs</code>	Lists all databases.
Insert Data	<code>db.<collection>.insertOne({field: "value"})</code>	Inserts a single document.
Find Data	<code>db.<collection>.find({})</code>	Queries all documents in a collection.

2. Express.js (Backend Framework)

Express.js is a minimal and flexible Node.js web application framework that provides a robust set of features for web and mobile applications.

Setup

To initialize a Node.js project and install Express:

1. Initialize project: `npm init -y`
2. Install Express: `npm install express`
3. Install Mongoose: `npm install mongoose`

Router Structure

A clean way to organize routes is using `express.Router()`.// Example in a file like 'routes/api.js'

```
const express = require('express');
```

```
const router = express.Router();
```

```
// Define a GET request endpoint
```

```
router.get('/', (req, res) => {
```

```
  res.send('API is running');
```

```
});
```

```
module.exports = router;
```

Middleware

Middleware functions are functions that have access to the request object (`req`), the response object (`res`), and the next middleware function in the application's request-response cycle.

- `app.use(express.json())`: Built-in middleware for parsing incoming JSON data.
- `app.use(express.urlencoded({ extended: true }))`: Built-in middleware for parsing incoming form data.

3. React (Frontend Library)

React is a JavaScript library for building user interfaces, primarily focused on component-based architecture.

Project Initialization

Create a new React application:

```
npx create-react-app <app-name>
```

Component Types

Type	Description	Example Hook
Functional Component	Defined as a JavaScript function; preferred for modern React development.	<code>useState</code> , <code>useEffect</code>
Class Component	Defined as an ES6 class; older method.	<code>this.state</code> , <code>componentDidMount</code>

State and Side Effects

- **`useState`**: Allows functional components to manage local state.
- **`useEffect`**: Handles side effects (like data fetching, subscriptions, or manually changing the DOM) after the component renders.

```
import React, { useState, useEffect } from 'react';

function ExampleComponent() {

  const [count, setCount] = useState(0);

  // Runs after every render

  useEffect(() => {

    document.title = `You clicked ${count} times`;

  }, [count]); // Dependency array: only re-run if 'count' changes

  return (

    <div>

      <p>You clicked {count} times</p>

      <button onClick={() => setCount(count + 1)}>

        Click me

      </button>

    </div>

  );

}
```

Communication

Component	Method	Description
Parent to Child	Props	Data passed down from parent to child components.
Child to Parent	Callback Functions	Child calls a function passed as a prop from the parent.

Component	Method	Description
Global State	Context API / Redux / Zustand	For managing state accessible across many components.

4. Node.js (Runtime Environment)

Node.js allows running JavaScript on the server side, acting as the foundation for Express.js.

Important Concepts

- **V8 Engine:** The high-performance JavaScript engine used by Chrome and Node.js.
- **Event Loop:** Node's core concept for handling asynchronous operations efficiently using a non-blocking I/O model.
- **NPM (Node Package Manager):** The world's largest software registry, used for managing dependencies.

Package Management

Command	Description
<code>npm install <package></code>	Installs a package locally as a dependency.
<code>npm install -g <package></code>	Installs a package globally.
<code>npm install --save-dev <package></code>	Installs a package as a development dependency (e.g., Linters, testing tools).

5. Deployment Checklist

Reviewing deployment specifics should be done before pushing code to a live server.

- **Environment Variables:** Store sensitive data (e.g., database URLs, API keys) in a `.env` file and exclude it from version control using File.

- **CORS:** Configure Cross-Origin Resource Sharing on the Express server to allow the React frontend to make requests.
- **Build Frontend:** Run `npm run build` in the React directory to create a production-ready static folder.
- **Server Host:** Choose a hosting provider for the backend (e.g., Heroku, Digital Ocean, AWS) and ensure the database (e.g., MongoDB Atlas) is accessible from the server's location Place.

For a deeper dive into the MERN stack deployment process, attend the next session: [Calendar event on Date with Person](#).

Class notes



Class notes

Owner Person

Instructor Person

Subject Psychology

Date

Lecture #1: Topic

- Jot down notes during class—develop a shorthand system that works for you
- You can underline and highlight with different colors to emphasize important information
- You can organize resources, questions, and key terms in the tabs

Date

Lecture #2: Topic

- Jot down notes during class—develop a shorthand system that works for you
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Important information

Resources

Name	Type	Notes
File	Textbook	Add page numbers and key chapters
File	Video	Add timestamps to focus on key sections
File	Article	Highlight key sections
Link	Website	Bookmark key pages

Key terms and *concepts*

Term	Definition	Notes
Word	Describe in 1-2 sentences	Add visuals, charts, supporting ideas, or relevant resources such as external links
Name	Describe in 1-2 sentences	The content should support your understanding of the term
Concept	Describe in 1-2 sentences	Memory devices or visual aids can help with recall

Questions and *discussion*

Question	Notes
Add question	• Add answers or important ideas from class discussion
Add question	• Be concise and clear

Homework

Homework

Task	Status	Due	Notes
Task 1	Submitted	Date	Note relevant details like formatting requirements
Task 2	In progress	Date	Break down bigger projects into smaller tasks
Task 3	Not started	Date	Reach out to your instructor if you're stuck or need an extension